

PROTOCOL: Tissue Sampling

v1.0

This protocol applies only to ‘large’ specimens—those that are pinned or in matrix tubes. Ensure your arrays are complete before proceeding. If your specimens are not labelled, ensure they remain in the same positions in the array.

(1) Prepare tissue sampling workstation

- § *Working with a ‘plate stand’, that elevates and angles the DNA extraction plate, can make this workflow more comfortable.*
- a. Obtain your DNA extraction plate and using a multi-channel pipette and reservoir, fill it with *either* **50–100 µL** EtOH (if extracting at a later date) or **50 µL** Alakline Lysis buffer (if extracting immediately).
- b. Ensure the correct box of specimens for the plate is at your workstation
- c. Obtain two pairs of forceps
- d. Fill a small jar with enough EtOH to fully cover the tips of your forceps; this can be Waste EtOH. A small, stout, jar is recommended to reduce the likelihood of spills.
- e. Set up an Ethanol burner on the periphery of your workstation, light it, then flame-sterilize your forceps.

(2) Sample tissue into plates

- § *For Alkaline lysis, we find that ‘less is best’ when it comes to tissue. The amount of tissue should be no larger than one-half of a single grain of rice.*
- § *We find it is helpful to ‘rotate’ the specimens 90° after sampling tissue to help the user follow along.*
- a. Light your EtOH burner
- b. Flame-sterilize forceps by dipping into the ethanol and then burning it off with the burner
- c. Obtain the specimen with tissue bound for well A01
- d. Remove the tip of a leg, or—if necessary—a small amount of body tissue and place into the well, ensuring it is fully covered by the buffer.
- e. Repeat b–d until you sample tissue into H11.
- f. Seal plate with a PCR seal if it contains AL buffer and will be incubated, or seal plate with a foil seal if it contains EtOH and will be stored.
- g. Proceed go DNA extraction.

